Dodecahedral Transport, Sector Geometry on L(Petersen), and the Identity $MM^{\mathsf{T}} = A^3 + 2A^2 + 2I$

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March 2026

Preprint DOI: 10.5281/zenodo.19043356

Abstract

We study a local transport system on the flags of the combinatorial dodecahedron and ask what exact incidence geometry survives on its quotient core. The construction yields a connected 60-vertex tetravalent chamber graph

$$G_{60} \cong \mathrm{AT4val}[60, 6],$$

together with a quotient tower

$$G_{60} \to G_{30} \to G_{15}, \qquad G_{15} \cong L(\text{Petersen}).$$

On the core graph G_{15} we define a transport-induced sector system and its 15×30 sector–edge incidence matrix M. The central result is that the corresponding Gram matrix

$$Q := MM^{\mathsf{T}}$$

is exactly the cubic polynomial

$$Q = A^3 + 2A^2 + 2I,$$

where A is the adjacency matrix of G_{15}. Equivalently, the sector overlaps are completely determined by graph distance in L(Petersen). After centering and normalization, the sector vectors yield a rigid three-angle spherical configuration in a 14-dimensional space. Alongside this metric structure, the intermediate quotient $G_{30} \to G_{15}$ carries a signed 2-lift and an associated non-trivial $\mathbb{Z}_2$-valued cocycle, giving a companion invariant not detected by the quadratic overlap geometry.

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1 Introduction

Local transport systems on highly symmetric combinatorial objects often produce global structure far more rigid than their local definitions initially suggest. This paper studies such a phenomenon on the flag structure of the combinatorial dodecahedron. The present preprint is archived at Zenodo [1]. From elementary local transport moves we obtain a connected 60-vertex tetravalent chamber graph G_{60}, which we call the *Thalean graph*. This graph admits a natural quotient tower

$$G_{60} \to G_{30} \to G_{15},$$

and the deepest quotient is the line graph of the Petersen graph:

$$G_{15} \cong L(\text{Petersen}).$$

This quotient provides the algebraic setting in which the deeper structure of the transport system becomes visible.

The first main layer of the paper is incidence-geometric. Each vertex of the core graph determines a transport sector, and these sectors assemble into a 15×30 sector–edge incidence matrix

$$M.$$

From this we form the central quadratic object of the paper,

$$Q := MM^{\mathsf{T}},$$

which we call the *Thalean quadratic*. The entries of Q record pairwise sector overlaps, so Q is the Gram matrix of the sector system. The main theorem identifies this quadratic object with a cubic polynomial in the adjacency operator of the core:

$$Q = MM^{\mathsf{T}} = A^3 + 2A^2 + 2I.$$

Equivalently, the quadratic geometry of the sectors is completely determined by the cubic polynomial

$$p(x) = x^3 + 2x^2 + 2$$

applied to the adjacency matrix of $L(\text{Petersen})$.

The second main layer is metric. The sector system has strong regularity: each sector has size 14, and each edge of the core lies in exactly 7 sectors. After subtracting the common mean vector and normalizing, the 15 sector vectors lie in the 14-dimensional subspace orthogonal to the all-ones direction and form a spherical configuration with exactly three distinct nontrivial inner products. More precisely, if $u \neq v$ are vertices of G_{15}, then

$$\langle x_u, x_v \rangle = \begin{cases} \frac{37}{112}, & d(u,v) = 1, \\ -\frac{23}{112}, & d(u,v) = 2, \\ -\frac{38}{112}, & d(u,v) = 3. \end{cases}$$

Thus the spherical geometry is completely determined by graph distance in the core. The sphere is not imposed externally; it is the normalized Euclidean realization of the quadratic overlap law.

A third layer persists beyond this quadratic and spherical picture. The intermediate quotient

$$G_{30} \to G_{15}$$

is a signed 2-lift, and the induced edge-signing defines a $\mathbb{Z}_2$-valued cocycle on G_{15}. This cocycle determines a nontrivial class in

$$H^1(G_{15}, \mathbb{Z}_2),$$

is not realizable by any local rule, and in the current presentation admits a minimal support representative of size 6. It is not detected by the quadratic sector observables. The construction therefore carries two complementary invariant layers: a metric layer encoded by the Thalean quadratic and its spherical realization, and a topological layer encoded by the cocycle and its loop holonomy.

We use the following terminology throughout. The overall local transport procedure on the dodecahedral flag structure is the *Thalean construction*. The resulting 60-vertex chamber graph is the *Thalean graph*. The distinguished connective incidence law induced on the core graph will be called the *thalion*. Within this language, the matrix M encodes the sector data of the thalion, and the Thalean quadratic

$$Q = MM^{\mathsf{T}}$$

records its overlap geometry.

The contribution of the paper is therefore structural. A local transport rule on the combinatorial dodecahedron compresses to a Petersen line-graph core, induces a quadratic sector geometry governed by a cubic adjacency law, realizes that law as a rigid three-angle spherical configuration, and still retains a nonlocal cocycle invisible to the quadratic sector observables.

The paper is organized as follows. Section 2 defines the Thalean construction and the resulting chamber graph. Section 3 describes the quotient tower and identifies the Petersen line-graph core. Section 4 introduces the thalion and its sector incidence matrix. Section 5 defines the Thalean quadratic, and Section 6 proves the polynomial incidence identity

$$Q = A^3 + 2A^2 + 2I.$$

Section 7 derives the spherical realization. Section 8 studies the signed lift and cocycle. The remaining sections synthesize the metric and topological aspects of the construction.

2 The Thalean Construction

We now describe the local transport procedure that generates the primary object of the paper. The input is the flag structure of the combinatorial dodecahedron. The output is a finite chamber system whose opposite-edge quotient is a connected 60-vertex tetravalent graph. This graph will be the Thalean graph.

The guiding point is that the construction is local. One does not begin with a preselected graph on 60 vertices. Instead, one begins with elementary incidence data on the dodecahedron and a small set of admissible transport moves. The global graph arises by closing these moves under iteration and then passing to the relevant quotient.

2.1 Flags and chamber states

Let $\mathcal{D}$ denote the combinatorial dodecahedron. We work with its flag structure, or equivalently with its chamber structure. A flag is a mutually incident triple consisting of a vertex, an edge, and a face of $\mathcal{D}$. The total number of such flags is

$$120.$$

These flags form the most refined local states of the construction. However, it is more convenient for the transport system to pass immediately to a coarser state space determined by opposite-edge identification. This produces the 60-state chamber system from which the Thalean graph is built.

2.2 Local transport moves

The Thalean construction is generated by elementary local moves on flags. These moves encode how one is permitted to pass from one incident configuration to a neighboring one while remaining inside the combinatorial structure of the dodecahedron.

At the formal level, the construction uses two distinguished transport types: face transport, which moves locally across a face-adjacent incidence, and vertex transport, which moves locally across a vertex-adjacent incidence.

These moves generate a finite transport system on the set of local states. The important point is that they are induced directly by the incidence relations internal to the dodecahedron. The resulting transport is therefore not an externally imposed graph dynamic, but a chamber system arising from the local combinatorics of the polyhedron.

2.3 Opposite-edge quotient

The raw flag system contains a natural redundancy. Opposite local configurations along an edge determine equivalent transport states for the purposes of the construction. We therefore pass from the 120 flags to a 60-element quotient by identifying opposite-edge partner states.

Definition 2.1 (Thalean chamber system)**.** *The* Thalean chamber system *is the* 60*-state quotient of the flag transport system on the combinatorial dodecahedron obtained by opposite-edge identification.*

This quotient is intrinsic to the transport picture developed here. It is not an auxiliary simplification introduced for convenience, but the first compression step in the construction.

2.4 The transport graph

We now form a graph on the 60 chamber states by declaring two states adjacent whenever one is obtained from the other by a single admissible transport move. Because the construction is generated by local face and vertex transport, each state has four neighbors in the resulting graph.

Definition 2.2 (Thalean graph)**.** *The* Thalean graph *is the graph whose vertices are the chamber states of the Thalean chamber system, with adjacency given by single-step admissible transport.*

This graph is therefore a transport graph rather than a graph chosen in advance. Its vertices are equivalence classes of local dodecahedral states, and its edges record the admissible one-step transitions between them.

2.5 Conceptual role of the construction

The Thalean construction should be viewed as a local-to-global mechanism. Locally, it consists only of finite incidence data and elementary admissible moves on dodecahedral flags. Globally, it produces a highly organized transport graph whose deeper quotient and sector geometry carry substantially richer structure.

At this stage, the main point is not yet the special identity satisfied by the sector matrix, nor the spherical realization, nor the cocycle. It is that all of those later structures arise from the same finite local input.

We summarize the role of the present section as follows:

$$\text{dodecahedral flags} \longrightarrow \text{local transport moves} \longrightarrow \text{opposite-edge quotient} \longrightarrow \text{Thalean graph.}$$

The next section studies the resulting graph as a concrete combinatorial object and identifies its canonical 15-vertex core.

3 The Thalean Graph and its Core

We now pass from the generative transport procedure to the resulting finite graphs. The first visible object is the 60-vertex chamber graph produced by the Thalean construction. This graph is already highly structured: it is connected, tetravalent, arc-transitive, has diameter 6, and contains nontrivial short-cycle data. More importantly, it admits a natural quotient tower

$$G_{60} \longrightarrow G_{30} \longrightarrow G_{15},$$

whose deepest term is the line graph of the Petersen graph.

3.1 The 60-vertex graph

Let G_{60} denote the graph obtained from the 60 chamber states of the Thalean construction by joining two chamber states whenever they are related by a single admissible transport move. This is the primary transport graph of the paper, and we identify it with the Thalean graph.

Proposition 3.1. *The graph G_{60} is connected and 4-regular.*

Proof. Each chamber state admits four elementary transport moves, so every vertex has degree 4. Connectedness follows from the connectedness of the underlying flag transport system and is verified computationally. □

The graph G_{60} has the invariant package

$$|V| = 60, \qquad |E| = 120, \qquad |\mathrm{Aut}(G)| = 480, \qquad \mathrm{diam}(G) = 6,$$

with shell vector

$$(1, 4, 8, 16, 24, 6, 1).$$

Theorem 3.2. *The graph G_{60} is isomorphic to the arc-transitive tetravalent graph*

$$\mathrm{AT4val}[60, 6],$$

as identified by Potočnik [3], and corresponds to `Graph52002` *in the House of Graphs [2].*

Proof. The transport construction yields a connected 4-regular graph on 60 vertices with the invariant package above. Comparison with the arc-transitive census and canonical encodings identifies it as $\mathrm{AT4val}[60, 6]$. □

The point of this identification is not that the graph is newly discovered, but that the Thalean construction recovers a known census graph together with additional quotient, incidence, and cocycle structure not visible in the census label alone.

3.2 The quotient tower

The chamber graph admits two natural quotient steps:

$$G_{60} \longrightarrow G_{30} \longrightarrow G_{15}.$$

The first quotient arises from the double-cover structure in the chamber graph; the second is the deeper quotient relevant to the incidence and cocycle structure. In the present paper, the crucial endpoint is G_{15}.

The quotient tower is best understood as an internal compression of the transport system. One begins with local chamber dynamics on G_{60}, passes through the intermediate 30-vertex layer, and reaches a 15-vertex graph on which the algebraic and geometric structure of the construction becomes fully visible.

3.3 The Petersen core

Theorem 3.3 (Petersen core)**.** *The graph G_{15} obtained as the deepest quotient of the Thalean construction is isomorphic to the line graph of the Petersen graph:*

$$G_{15} \cong L(\mathrm{Petersen}).$$

Proof. The quotient construction yields a 15-vertex 4-regular graph with 30 edges. Explicit graph-isomorphism verification identifies it with $L(\mathrm{Petersen})$. □

Under this identification, the vertices of G_{15} may be regarded as the 15 edges of the Petersen graph, and adjacency in G_{15} records incidence of those edges at a common endpoint.

3.4 Why the core matters

The appearance of $L(\text{Petersen})$ is the structural hinge of the paper. The 60-vertex graph is the transport object, but the 15-vertex quotient is the algebraic core. It is on G_{15} that the incidence matrix M is defined, the Thalean quadratic

$$Q := MM^{\mathsf{T}}$$

is formed, and the cubic identity

$$Q = A^3 + 2A^2 + 2I$$

is proved.

The passage to G_{15} is therefore not a loss of structure in any naive sense. Rather, it is the level at which the transport system becomes algebraically legible.

3.5 Summary

The outcome of the present section may be summarized as

$$\text{Thalean construction} \longrightarrow G_{60} \longrightarrow G_{30} \longrightarrow G_{15} \cong L(\text{Petersen}).$$

The graph G_{60} is the full chamber transport graph. The graph G_{15} is its canonical compressed core. The next section introduces the thalion and the sector–edge incidence matrix on this core.

4 Thalion and Sector Incidence

We now introduce the connective incidence structure carried by the core graph and the finite incidence data derived from it. This is the point at which the transport construction becomes a concrete algebraic object.

4.1 The thalion

The 15-vertex core

$$G_{15} \cong L(\text{Petersen})$$

is not used here as a bare graph in isolation. It is equipped with the connective law inherited from the Thalean construction: a structured family of local relations determining which incidences are admitted, how sectors are organized, and how transport data is assembled into the global core.

Definition 4.1 (Thalion)**.** *The* thalion *of a Thalean construction is its distinguished connective incidence law: the organized system of admissible local relations through which the construction coheres and through which its transport structure descends to the core graph.*

The thalion is therefore more informative than adjacency on G_{15} alone. It records not just which core vertices are adjacent, but how the transport system organizes the local relation pattern from which the sector structure arises.

4.2 Sectors on the core

Each vertex of G_{15} determines a distinguished collection of edges of G_{15}, defined by the transport geometry inherited from the construction. We denote the sector associated with a vertex $v \in V(G_{15})$ by

$$S(v) \subseteq E(G_{15}).$$

The sectors are not chosen ad hoc. They are induced by the thalion. Each sector records the edge content locally visible from a given core vertex under the transport incidence law. This induced sector system is the first concrete finite realization of the connective structure carried by the core.

4.3 The sector–edge incidence matrix

Definition 4.2 (Sector–edge incidence matrix)**.** *Let the rows of M be indexed by the vertices of G_{15}, and let the columns of M be indexed by the edges of G_{15}. The* sector–edge incidence matrix *is the matrix*

$$M = (M_{v,e}) \in \{0,1\}^{15\times 30}$$

defined by

$$M_{v,e} = \begin{cases} 1, & e \in S(v), \\ 0, & e \notin S(v). \end{cases}$$

Thus each row of M is the characteristic vector of a sector, expressed in edge coordinates.

4.4 Interpretation of rows and columns

Rows correspond to vertices of the core graph, or equivalently to sectors. Columns correspond to edges of the core graph. The entry $M_{v,e}$ records whether the edge e lies in the sector based at v.

This gives two compatible ways to read the matrix. First, one may regard M combinatorially, as a finite incidence table. Second, one may regard M geometrically, as a family of vectors in $\mathbb{R}^{30}$. In the latter view, the sector system becomes a configuration of vectors whose overlap structure will be encoded by the Gram matrix MM^{T}.

4.5 Regularity of the sector system

The sector system has strong internal regularity. Each row of M has the same weight, since every sector has the same cardinality, and each edge lies in the same number of sectors. In the present construction, each sector has size 14, and each edge lies in exactly 7 sectors.

This uniformity is one of the key reasons the sector system is useful. It means that the thalion induces not merely a family of subsets, but a balanced incidence geometry. That balance is what later allows the overlap matrix to be expressed as a polynomial in the adjacency operator of the core graph.

4.6 The role of M

The matrix M is the bridge between transport and geometry. It is built from the thalion, but it already points beyond pure incidence theory.

- From the transport point of view, M records the finite sector data induced on the core.

- From the graph-theoretic point of view, M is a structured vertex-by-edge incidence matrix on G_{15}.
- From the metric point of view, M is a family of vectors whose Gram matrix measures sector overlap.
- From the algebraic point of view, M is the object from which the Thalean quadratic is formed.

This multiplicity of interpretations is exactly what makes the sector matrix central. It is the place where the combinatorial, geometric, and algebraic aspects of the construction first meet.

4.7 Summary

The outcome of this section is the passage

$$\text{thalion} \longrightarrow \text{sector system} \longrightarrow M.$$

The thalion gives the connective law on the core, the sectors encode that law as edge subsets, and the matrix M packages those sectors into a concrete algebraic object. The next section defines the Thalean quadratic

$$Q := MM^{\mathsf{T}},$$

which records the sector-overlap geometry.

5 The Thalean Quadratic

With the sector–edge incidence matrix M now in place, the next object is forced upon us. The rows of M are vectors in $\mathbb{R}^{30}$, one for each sector of the core graph G_{15}. The natural quadratic object attached to such a family of vectors is its Gram matrix.

Definition 5.1 (Thalean quadratic)**.** *Let M be the sector–edge incidence matrix of the Thalean construction on the core graph G_{15}. The* Thalean quadratic *is the matrix*

$$Q := MM^{\mathsf{T}}.$$

Since M is a 15×30 matrix, the Thalean quadratic Q is a 15×15 symmetric matrix indexed by the vertices of G_{15}. Its (u, v)-entry is

$$Q_{uv} = (MM^{\mathsf{T}})_{uv} = \sum_{e \in E(G_{15})} M_{u,e} M_{v,e}.$$

Proposition 5.2. *For vertices $u, v \in V(G_{15})$,*

$$Q_{uv} = |S(u) \cap S(v)|.$$

Proof. By definition,

$$Q_{uv} = \sum_{e} M_{u,e} M_{v,e}.$$

For a given edge e, the product $M_{u,e}M_{v,e}$ is equal to 1 if and only if $e \in S(u) \cap S(v)$, and is 0 otherwise. Summing over all edges counts the common elements of the two sectors. □

Thus Q is the sector-overlap matrix of the thalion.

5.1 Why this is the right quadratic object

The matrix $Q = MM^{\mathsf{T}}$ is the canonical quadratic form associated with the sector system. As a Gram matrix, it records the inner products of the sector vectors. As an overlap matrix, it records the intersection sizes of sectors. In this sense, it is the first metric object naturally induced by the transport data.

In particular, the diagonal entries record sector size:

$$Q_{uu} = |S(u)| = 14.$$

The off-diagonal entries record pairwise overlap:

$$Q_{uv} = |S(u) \cap S(v)|, \qquad u \neq v.$$

The passage

$$M \longmapsto Q = MM^{\mathsf{T}}$$

forgets the individual edge coordinates and retains only pairwise overlap data. This is precisely why Q is the right object to study next: it compresses the incidence structure of the thalion into a symmetric geometric form.

5.2 Distance-based interpretation

Since $G_{15} \cong L(\text{Petersen})$ has diameter 3, any two vertices u, v lie at one of four possible graph distances:

$$d(u, v) \in \{0, 1, 2, 3\}.$$

The overlap matrix Q will therefore be describable by four corresponding overlap values. Equivalently, Q can be expanded in the distance-matrix basis

$$Q = \alpha_0 D_0 + \alpha_1 D_1 + \alpha_2 D_2 + \alpha_3 D_3,$$

where D_i is the matrix indicating vertex pairs at distance i.

This already suggests that the quadratic geometry of the sectors is controlled by the distance-regular structure of the core. The next section makes that control explicit by identifying Q with a cubic polynomial in the adjacency matrix A.

5.3 Symmetry and regularity

Because Q is a Gram matrix, it is automatically symmetric:

$$Q^{\mathsf{T}} = Q.$$

Moreover, the regularity of the sector system implies that its diagonal entries are constant, and the symmetry of the core graph strongly constrains the off-diagonal entries. The overlap geometry is therefore far from arbitrary.

This is the first clear indication that Q should lie in the adjacency algebra of the core graph. The main theorem of the next section confirms exactly that: the Thalean quadratic is not merely compatible with the adjacency algebra of G_{15}, but is given by a specific cubic polynomial in A.

5.4 Framing identity

The central object of the paper may now be framed in its simplest form:

$$\boxed{Q := MM^{\mathsf{T}}.}$$

This identity should be read as the definition of the Thalean quadratic. It says that the paper's main quadratic invariant is the Gram matrix of the sector system.

The next section upgrades this definition to a structural theorem:

$$\boxed{Q = MM^{\mathsf{T}} = A^3 + 2A^2 + 2I.}$$

5.5 Summary

The purpose of the present section has been to identify the correct quadratic object attached to the Thalean construction. Starting from the sector–edge incidence matrix M, we formed the Gram matrix

$$Q := MM^{\mathsf{T}},$$

interpreted its entries as sector overlaps, and explained why it is the natural metric compression of the thalion. The next section proves that this quadratic object is governed exactly by a cubic polynomial in the adjacency operator of $G_{15} \cong L(\text{Petersen})$.

6 Sector Overlaps and the Polynomial Incidence Identity

This section contains the central theorem of the paper. The sector–edge incidence matrix M defined on the quotient core

$$G_{15} \cong L(\text{Petersen})$$

determines the quadratic overlap matrix

$$Q := MM^{\mathsf{T}}.$$

We show first that the entries of Q depend only on graph distance in G_{15}, and then identify the resulting distance expansion with a cubic polynomial in the adjacency matrix.

6.1 The main identity

Let A denote the adjacency matrix of $G_{15} \cong L(\text{Petersen})$.

Theorem 6.1 (Polynomial Incidence Identity)**.** *Let M be the sector–edge incidence matrix on G_{15}, and let*

$$Q := MM^{\mathsf{T}}.$$

Then

$$\boxed{Q = A^3 + 2A^2 + 2I.}$$

Equivalently,

$$\boxed{MM^{\mathsf{T}} = A^3 + 2A^2 + 2I.}$$

The proof is organized in three steps. First, we compute the sector overlaps by distance class. Second, we rewrite the overlap matrix in the distance-matrix basis. Third, we reduce that distance expansion explicitly inside the adjacency algebra of G_{15}.

6.2 Sector overlaps by distance class

For vertices $u, v \in V(G_{15})$, the entry Q_{uv} counts the overlap of the sectors based at u and v:

$$Q_{uv} = |S(u) \cap S(v)|.$$

Since $G_{15} \cong L(\text{Petersen})$ has diameter 3, the possible distances are

$$d(u, v) \in \{0, 1, 2, 3\}.$$

Proposition 6.2 (Distance-overlap formula)**.** *For all vertices $u, v \in V(G_{15})$,*

$$Q_{uv} = \begin{cases} 14, & d(u,v) = 0, \\ 9, & d(u,v) = 1, \\ 5, & d(u,v) = 2, \\ 4, & d(u,v) = 3. \end{cases}$$

Proof. The diagonal value is immediate:

$$Q_{uu} = |S(u)| = 14,$$

since every sector has size 14.

For $u \neq v$, direct enumeration of the sector supports shows that the overlap $|S(u) \cap S(v)|$ depends only on the graph distance $d(u, v)$, and the corresponding values are

$$|S(u) \cap S(v)| = \begin{cases} 9, & d(u,v) = 1, \\ 5, & d(u,v) = 2, \\ 4, & d(u,v) = 3. \end{cases}$$

This gives the stated formula. □

Remark 6.3. *The content of Proposition 6.2 is not merely the numerical values $9, 5, 4$, but the fact that the overlap matrix is controlled entirely by graph distance in the quotient core. This places the quadratic overlap geometry inside the distance algebra of G_{15}.*

6.3 Distance-matrix expansion

Let D_0, D_1, D_2, D_3 denote the distance matrices of G_{15}, so that

$$(D_i)_{uv} = \begin{cases} 1, & d(u,v) = i, \\ 0, & d(u,v) \neq i. \end{cases}$$

Then $D_0 = I$ and $D_1 = A$. Proposition 6.2 gives the following expansion.

Corollary 6.4 (Distance expansion of Q)**.** *The overlap matrix satisfies*

$$Q = 14D_0 + 9D_1 + 5D_2 + 4D_3.$$

Equivalently,

$$MM^{\mathsf{T}} = 14D_0 + 9D_1 + 5D_2 + 4D_3.$$

This formula is the structural bridge between the sector combinatorics and the adjacency algebra of the core graph.

6.4 Explicit reduction in the adjacency algebra of $L(\text{Petersen})$

Because $G_{15} \cong L(\text{Petersen})$ is distance-regular of diameter 3, its distance matrices lie in the adjacency algebra generated by A. We now write the distance matrices explicitly in terms of A.

Since G_{15} is 4-regular and every adjacent pair has exactly one common neighbor, one has

$$A^2 = 4D_0 + D_1 + D_2.$$

Using $D_0 = I$ and $D_1 = A$, this gives

$$D_2 = A^2 - A - 4I.$$

Next, because G_{15} is connected and regular on 15 vertices, the all-ones matrix J is the projection onto the 4-eigenspace of A. Using the spectrum

$$\{-2^{(5)},\, -1^{(4)},\, 2^{(5)},\, 4^{(1)}\},$$

one obtains

$$J = \frac{1}{4}(A - 2I)(A + I)(A + 2I) = \frac{1}{4}(A^3 + A^2 - 4A - 4I).$$

Since

$$J = D_0 + D_1 + D_2 + D_3,$$

it follows that

$$D_3 = J - I - A - D_2.$$

Substituting the formula for J and the expression for D_2 yields

$$D_3 = \frac{1}{4}(A^3 + A^2 - 4A - 4I) - I - A - (A^2 - A - 4I),$$

hence

$$D_3 = \frac{1}{4}(A^3 - 3A^2 - 4A + 8I).$$

We may now reduce the distance expansion explicitly:

$$\begin{aligned} Q &= 14D_0 + 9D_1 + 5D_2 + 4D_3 \\ &= 14I + 9A + 5(A^2 - A - 4I) + 4 \cdot \frac{1}{4}(A^3 - 3A^2 - 4A + 8I) \\ &= 14I + 9A + 5A^2 - 5A - 20I + A^3 - 3A^2 - 4A + 8I \\ &= A^3 + 2A^2 + 2I. \end{aligned}$$

This is exactly the claimed polynomial form.

6.5 Proof of the polynomial form

Proof of Theorem 6.1. By Corollary 6.4,

$$Q = 14D_0 + 9D_1 + 5D_2 + 4D_3.$$

The explicit adjacency-algebra reduction above shows that

$$14D_0 + 9D_1 + 5D_2 + 4D_3 = A^3 + 2A^2 + 2I.$$

Therefore

$$Q = A^3 + 2A^2 + 2I.$$

Equivalently,

$$MM^{\mathsf{T}} = A^3 + 2A^2 + 2I.$$

□

6.6 Spectral check

As a consistency check, recall that the spectrum of

$$G_{15} \cong L(\text{Petersen})$$

is

$$\{-2^{(5)},\, -1^{(4)},\, 2^{(5)},\, 4^{(1)}\}.$$

Applying the polynomial

$$p(x) = x^3 + 2x^2 + 2$$

to these eigenvalues gives

$$p(-2) = 2, \qquad p(-1) = 3, \qquad p(2) = 18, \qquad p(4) = 98.$$

These are precisely the eigenvalues of $Q = MM^{\mathsf{T}}$, so the polynomial form is consistent with the computed overlap matrix.

The spectral calculation is not the conceptual heart of the theorem; it is a secondary verification. The structural content is the combination of the distance-overlap formula and the explicit adjacency-algebra reduction.

6.7 Interpretation

The theorem says that the transport-induced sector geometry on the quotient core is not arbitrary. The overlap matrix of the sectors is completely determined by a cubic polynomial in the adjacency operator of $L(\text{Petersen})$. In particular, the transport construction induces a concrete incidence geometry on the core whose quadratic form lies in the adjacency algebra and is given by the specific polynomial

$$x^3 + 2x^2 + 2.$$

This is the central mathematical point of the paper. It is not merely that the transport construction reaches a known graph, but that the induced sector system on that graph satisfies an exact and rigid algebraic law.

6.8 Summary

The argument of this section has four steps:

1. define the overlap matrix by
$$Q_{uv} = |S(u) \cap S(v)|;$$
2. compute the sector overlaps by graph distance;
3. rewrite the result as
$$Q = 14D_0 + 9D_1 + 5D_2 + 4D_3;$$
4. reduce this distance expansion to the cubic identity
$$Q = A^3 + 2A^2 + 2I.$$

The next section turns this quadratic identity into geometry by centering and normalizing the sector vectors, thereby obtaining the induced three-angle spherical realization.

7 The Induced Three-Angle Spherical Realization

Once the sector Gram matrix has been identified, the corresponding spherical geometry follows naturally. The purpose of this section is to pass from the quadratic overlap structure

$$Q = MM^{\mathsf{T}}$$

to a normalized Euclidean configuration. The result is a three-angle spherical code associated with the transport-induced sector system on

$$G_{15} \cong L(\text{Petersen}).$$

7.1 Centered sector vectors

Let $m_v \in \mathbb{R}^{30}$ denote the row of M corresponding to the vertex $v \in V(G_{15})$. Since each sector has the same cardinality, all row vectors have the same coordinate sum. We therefore remove the trivial all-ones component by subtracting the mean vector

$$\mu := \frac{1}{15} \sum_{v \in V(G_{15})} m_v.$$

Define the centered sector vectors by

$$\widetilde{m}_v := m_v - \mu.$$

By construction,

$$\sum_{v \in V(G_{15})} \widetilde{m}_v = 0.$$

So the centered vectors lie in the hyperplane orthogonal to the all-ones direction.

Proposition 7.1. *The centered sector vectors lie in a subspace of dimension at most* 14.

Proof. The vectors $\widetilde{m}_v$ satisfy one linear relation,

$$\sum_{v \in V(G_{15})} \widetilde{m}_v = 0,$$

so they lie in a codimension-one subspace of the 15-dimensional row space. Equivalently, they lie in the hyperplane orthogonal to the all-ones vector. □

7.2 Normalized vectors

To obtain a spherical configuration, we normalize the centered vectors:

$$x_v := \frac{\widetilde{m}_v}{\|\widetilde{m}_v\|}.$$

Definition 7.2 (Induced spherical realization)**.** *The* induced spherical realization *of the transport sector system is the set*

$$\mathcal{X} := \{x_v\}_{v \in V(G_{15})}$$

of normalized centered sector vectors.

This is a derived object: it is obtained from the transport-induced sector matrix M by centering and normalization. The sphere is therefore not an independent input, but a geometric consequence of the incidence law.

7.3 Inner products from the overlap matrix

Because $Q = MM^{\mathsf{T}}$, all inner products among the uncentered row vectors are already known. Centering subtracts the trivial component, and normalization converts the centered Gram matrix into unit-vector inner products. The resulting values depend only on graph distance in the core.

Theorem 7.3 (Three-angle spherical realization)**.** *Let $\{x_v\}_{v\in V(G_{15})}$ be the normalized centered sector vectors. Then $\mathcal{X}$ is a spherical code in a 14-dimensional space, and for distinct vertices $u, v \in V(G_{15})$,*

$$\langle x_u, x_v\rangle = \begin{cases} \frac{37}{112}, & d(u,v) = 1, \\ -\frac{23}{112}, & d(u,v) = 2, \\ -\frac{38}{112}, & d(u,v) = 3. \end{cases}$$

Thus the normalized sector geometry is completely determined by graph distance in $G_{15} \cong L(\text{Petersen})$. The spherical code has exactly three nontrivial inner products, corresponding to the three nonzero distance classes.

7.4 Derivation from the quadratic identity

The point of Theorem 7.3 is not that a spherical picture can be drawn, but that the spherical realization is forced by the quadratic overlap law. The logical chain is:

transport construction $\longrightarrow$ sector matrix $M \longrightarrow Q = MM^{\mathsf{T}} \longrightarrow$ centered Gram matrix $\longrightarrow$ spherical code.

Since Section 6 proves

$$Q = A^3 + 2A^2 + 2I,$$

the spherical configuration may be regarded as the normalized Euclidean form of that cubic incidence law.

7.5 Geometric meaning

The induced spherical realization gives a compact geometric encoding of the sector overlaps. Several points are worth emphasizing.

- It is canonical: it is determined by the sector matrix M.
- It is distance-controlled: inner products depend only on graph distance in the quotient core.
- It is three-angular: exactly three nontrivial pairwise inner products occur.
- It is derived from the transport-induced incidence structure rather than introduced independently.

The spherical realization is therefore best viewed as a consequence of the main identity, not as a competing center of the paper.

7.6 What this geometry does not capture

Although the induced spherical configuration is rigid, it does not exhaust the transport structure. The signed lift

$$G_{30} \to G_{15}$$

carries a cocycle that is invisible to the quadratic overlap matrix and hence to the spherical realization derived from it. The sphere captures the metric side of the construction, but not the full lift data.

7.7 Summary

Starting from the sector matrix M, we center and normalize its row vectors to obtain a spherical code in a 14-dimensional space. The resulting configuration has exactly three nontrivial inner products,

$$\frac{37}{112}, \qquad -\frac{23}{112}, \qquad -\frac{38}{112},$$

determined by graph distance in $G_{15} \cong L(\text{Petersen})$. This is the geometric consequence of the polynomial incidence identity. The next section turns to the companion signed-lift invariant that is not detected by this quadratic structure.

8 The Signed Lift and a Companion Cocycle Invariant

The previous sections identify the transport-induced sector geometry on the quotient core and show that its Gram matrix satisfies the exact cubic identity

$$Q = MM^{\mathsf{T}} = A^3 + 2A^2 + 2I.$$

We now turn to a different invariant arising from the intermediate quotient

$$G_{30} \to G_{15}.$$

This quotient carries the structure of a signed 2-lift. The associated $\mathbb{Z}_2$-valued cocycle is not determined by the quadratic overlap matrix Q, and therefore records information that is complementary to the sector geometry.

8.1 The signed lift

The map

$$G_{30} \to G_{15}$$

is a double cover in which each vertex of G_{15} has a fiber of size 2. For a vertex $u \in V(G_{15})$, we write the two lifted vertices as

$$u^+, \qquad u^-.$$

Each edge of the base graph lifts in one of two ways:

- *parallel*:
$$u^+ \sim v^+, \qquad u^- \sim v^-;$$
- *crossed*:
$$u^+ \sim v^-, \qquad u^- \sim v^+.$$

This yields a sign function

$$\sigma : E(G_{15}) \to \{\pm 1\},$$

where

$$\sigma(e) = \begin{cases} +1, & \text{if } e \text{ lifts in parallel,} \\ -1, & \text{if } e \text{ lifts crossed.} \end{cases}$$

8.2 The cocycle

Passing from $\{\pm 1\}$ to $\mathbb{Z}_2$, define

$$\varepsilon(e) = \begin{cases} 0, & \sigma(e) = +1, \\ 1, & \sigma(e) = -1. \end{cases}$$

Definition 8.1 (Signed-lift cocycle)**.** *The* signed-lift cocycle *of the transport construction is the function*

$$\varepsilon : E(G_{15}) \to \mathbb{Z}_2$$

defined by the sign rule above.

This cocycle records whether transport across a given edge preserves the sheet or exchanges it. It is therefore the natural $\mathbb{Z}_2$-valued invariant attached to the signed lift.

8.3 Loop holonomy

If

$$C \subseteq G_{15}$$

is a cycle, define its holonomy parity by

$$\omega(C) := \sum_{e \in C} \varepsilon(e) \pmod 2.$$

Proposition 8.2 (Holonomy criterion)**.** *For any cycle $C \subseteq G_{15}$,*

$$\omega(C) = 0$$

if and only if the lifted walk closes on its original sheet, and

$$\omega(C) = 1$$

if and only if it closes on the opposite sheet.

Proof. Each crossed edge toggles the sheet, while each parallel edge preserves it. The total parity of crossed edges along the cycle therefore determines whether the lifted walk returns to its starting sheet or the opposite one. □

This is the key distinction between the cocycle and the overlap geometry. The matrix Q measures pairwise overlap of sector supports, whereas ε is detected only by transport around loops.

8.4 Switching equivalence and cohomology

The cocycle is intrinsic only up to switching at vertices. A switching function

$$\phi : V(G_{15}) \to \mathbb{Z}_2$$

acts on edge values by

$$\varepsilon'(uv) = \varepsilon(uv) + \phi(u) + \phi(v) \pmod 2.$$

Thus the correct invariant is the switching class of ε, or equivalently the cohomology class

$$[\varepsilon] \in H^1(G_{15}, \mathbb{Z}_2).$$

Theorem 8.3 (Cocycle nontriviality)**.** *The signed-lift cocycle defines a nontrivial cohomology class:*

$$[\varepsilon] \neq 0 \in H^1(G_{15}, \mathbb{Z}_2).$$

Proof. The cocycle is nontrivial because there exist cycles with odd holonomy. By Proposition 8.2, such a cycle cannot occur if ε were a coboundary. Hence $[\varepsilon] \neq 0$. □

In the verified data, the switching class admits a representative of minimal support size 6.

8.5 Failure of local determination

The cocycle is not recoverable from finite-radius local edge data on the core. This is because $G_{15} \cong L(\text{Petersen})$ is edge-transitive, so all rooted local edge neighborhoods are isomorphic, whereas the cocycle takes both values 0 and 1.

Theorem 8.4 (Failure of local determination)**.** *The signed-lift cocycle cannot be recovered from any finite-radius, automorphism-invariant local rule on edges of G_{15}.*

Proof. Any such local rule would assign the same value to every edge, since all local rooted edge neighborhoods are isomorphic. But the cocycle takes both values 0 and 1, so no such rule can recover it. □

This theorem should be read modestly. The point is not that graph cocycles are novel in themselves, but that for the present transport construction the signed-lift invariant is independent of the sector overlap geometry and is not visible in the local edge neighborhoods of the core graph.

8.6 Independence from the quadratic overlap structure

The role of the cocycle in this paper is now clear.

- The sector system yields the incidence matrix M, the overlap matrix

  $$Q = MM^{\mathsf{T}},$$

  the cubic identity

  $$Q = A^3 + 2A^2 + 2I,$$

  and the induced three-angle spherical realization.

- The signed lift yields the cocycle

  $$\varepsilon : E(G_{15}) \to \mathbb{Z}_2,$$

  its cohomology class $[\varepsilon]$, and the associated loop-holonomy data.

The cocycle is therefore a companion invariant. It is part of the structure induced by the transport construction, but it is not encoded by the quadratic matrix Q.

8.7 Summary

The intermediate quotient

$$G_{30} \to G_{15}$$

defines a signed 2-lift and a corresponding $\mathbb{Z}_2$-valued cocycle

$$\varepsilon : E(G_{15}) \to \mathbb{Z}_2.$$

This cocycle is detected by loop holonomy, is nontrivial in

$$H^1(G_{15}, \mathbb{Z}_2),$$

and cannot be recovered from local edge data on the core. Its role in the paper is not to compete with the main theorem, but to record a companion invariant of the transport construction that is independent of the quadratic overlap structure.

9 Discussion

The main contribution of the paper is an exact incidence theorem on the quotient core

$$G_{15} \cong L(\text{Petersen}).$$

Starting from local transport on dodecahedral flags, one obtains a sector system on G_{15}, a sector–edge incidence matrix M, and the quadratic overlap matrix

$$Q := MM^{\mathsf{T}}.$$

The central result is that this matrix is not arbitrary, but satisfies the exact polynomial identity

$$Q = A^3 + 2A^2 + 2I.$$

This is the structural point of the paper. The construction does more than produce a known finite graph: it induces on a canonical quotient core a specific sector geometry whose Gram matrix is governed by a cubic polynomial in the adjacency operator. The passage

$$\text{dodecahedral transport} \longrightarrow G_{60} \longrightarrow G_{15} \cong L(\text{Petersen}) \longrightarrow M \longrightarrow Q = A^3 + 2A^2 + 2I$$

is therefore the main mathematical throughline.

9.1 Why the theorem is not merely computational

One possible reaction to the identity

$$Q = A^3 + 2A^2 + 2I$$

is that it simply reflects the fact that $L(\text{Petersen})$ is distance-regular and that Q lies in its adjacency algebra. That is true as far as it goes, but it misses the main point. The content of the paper is not merely that some matrix in the construction belongs to the adjacency algebra of $L(\text{Petersen})$, but that the transport-induced sector system yields the specific distance expansion

$$Q = 14D_0 + 9D_1 + 5D_2 + 4D_3,$$

and hence the specific cubic polynomial

$$A^3 + 2A^2 + 2I.$$

So the theorem is both combinatorial and algebraic. Combinatorially, it arises from the sector overlaps induced by the transport construction. Algebraically, it collapses to a polynomial identity in the adjacency operator of the quotient core.

9.2 The role of the quotient core

The quotient

$$G_{15} \cong L(\text{Petersen})$$

is what makes the transport system algebraically legible. At the level of the full chamber graph G_{60}, one sees the transport dynamics and the finite closure of the construction. At the level of the quotient core, one can define the transport-induced sector system, compute its overlap matrix, and identify that matrix inside the adjacency algebra of a classical graph.

In this sense, the quotient is not a loss of mathematical content, but a clarification of it. The core is the level at which the exact incidence law becomes visible.

9.3 The induced spherical geometry

Once the quadratic identity is known, the spherical realization is natural. Centering and normalizing the row vectors of M yields a three-angle spherical code in a 14-dimensional space. This geometric realization is a consequence of the overlap law, not an independent starting point.

Its significance is that the transport-induced sector system admits a rigid Euclidean form with inner products determined entirely by graph distance in the core. This supplies a geometric shadow of the cubic incidence law.

9.4 The signed-lift cocycle

The signed-lift cocycle plays a different role. It is not part of the proof of the main theorem, and it is not encoded by the quadratic overlap matrix. Rather, it is a companion invariant associated with the intermediate double cover

$$G_{30} \to G_{15}.$$

The cocycle records loop-holonomy data that is invisible to Q. Its presence shows that the transport construction carries more information than the sector Gram matrix alone.

This point should be understood modestly. The paper does not claim that graph cocycles are novel in themselves. The claim is only that, for this particular transport construction, the signed-lift cocycle is an induced invariant that is independent of the quadratic overlap structure.

9.5 What the paper does and does not claim

The paper does not claim that the graph AT4val[60, 6] is new, nor that the line graph of the Petersen graph is being studied here for the first time. It also does not claim that every ingredient of the paper is equally central. The real contribution is the exact cubic incidence law on the quotient core.

More precisely, the paper claims the following:

1. a local transport rule on dodecahedral flags produces a finite chamber graph G_{60};
2. this graph has a quotient core
$$G_{15} \cong L(\text{Petersen});$$
3. the induced sector incidence matrix M on that core satisfies
$$MM^{\mathsf{T}} = A^3 + 2A^2 + 2I;$$
4. this yields a rigid three-angle spherical realization;

5. the intermediate signed lift carries a companion $\mathbb{Z}_2$-valued cocycle not detected by $MM^\top$.

That is the mathematical content of the paper in its most compact form.

9.6 Outlook

There are several natural directions for further work.

First, one can ask whether similar transport-induced sector systems arise on other quotient cores coming from regular polyhedra or chamber systems.

Second, one can ask whether the cubic polynomial

$$x^3 + 2x^2 + 2$$

admits a more intrinsic characterization from the transport side, rather than appearing only after passage to the quotient core.

Third, one can ask how general the separation is between the quadratic overlap geometry and companion signed-lift invariants of the kind studied here.

These questions lie beyond the scope of the present paper. What has been shown here is that one local transport construction already yields an exact and nontrivial incidence theorem on a classical quotient core.

9.7 Summary

The paper is best read as a theorem about induced sector geometry on

$$G_{15} \cong L(\text{Petersen}).$$

The transport construction provides the origin of the data, the quotient core provides the algebraic stage, and the identity

$$MM^\top = A^3 + 2A^2 + 2I$$

provides the main result. The spherical realization and the signed-lift cocycle record, respectively, the geometric and cohomological consequences that travel alongside this central incidence law.

10 Conclusion

We have studied a local transport rule on the flags of the combinatorial dodecahedron and asked what exact incidence geometry survives on its quotient core. The answer is that the construction yields a finite chamber graph

$$G_{60},$$

a quotient tower

$$G_{60} \to G_{30} \to G_{15},$$

and a canonical 15-vertex core

$$G_{15} \cong L(\text{Petersen})$$

on which the induced sector system satisfies an exact cubic overlap law.

More precisely, if M denotes the sector–edge incidence matrix on G_{15}, then its Gram matrix

$$Q := MM^\top$$

satisfies the identity

$$Q = A^3 + 2A^2 + 2I.$$

This is the central theorem of the paper. It shows that the transport-induced sector geometry on the quotient core is governed by a specific polynomial in the adjacency operator of $L(\text{Petersen})$.

Two further structures accompany this theorem. First, centering and normalizing the row vectors of M yields a rigid three-angle spherical configuration in a 14-dimensional space. Second, the intermediate quotient

$$G_{30} \to G_{15}$$

defines a signed 2-lift and an associated nontrivial $\mathbb{Z}_2$-valued cocycle. These record, respectively, a geometric consequence and a companion lift invariant of the transport construction.

The paper is therefore not simply about constructing a known graph. Its main point is that local dodecahedral transport induces, on a canonical quotient core, an exact and rigid incidence law:

$$MM^{\mathsf{T}} = A^3 + 2A^2 + 2I.$$

That identity is the mathematical center of the work.

A Computational Verification

This appendix records the finite computations supporting the main structural claims of the paper. Its purpose is reproducibility. The computational pipeline verifies the transport graph, the quotient tower, the quotient core $G_{15} \cong L(\text{Petersen})$, the sector matrix M, the overlap matrix

$$Q := MM^{\mathsf{T}},$$

the identity

$$Q = A^3 + 2A^2 + 2I,$$

the induced spherical realization, and the signed-lift cocycle.

A.1 Verified claims

The computations support the following statements.

1. The Thalean construction produces a connected 60-vertex tetravalent graph G_{60}.
2. The graph G_{60} satisfies

   $$|V| = 60, \qquad |E| = 120, \qquad \deg(v) = 4, \qquad \operatorname{diam}(G_{60}) = 6,$$

   with shell vector

   $$(1, 4, 8, 16, 24, 6, 1),$$

   triangle count 40, and automorphism group order 480.
3. The graph G_{60} is isomorphic to the census graph

   $$\text{AT4val}[60, 6],$$

   corresponding to `Graph52002` in the House of Graphs [3, 2].

4. The quotient tower
$$G_{60} \longrightarrow G_{30} \longrightarrow G_{15}$$
is correctly constructed.

5. The deepest quotient satisfies
$$G_{15} \cong L(\text{Petersen}).$$

6. The transport-induced sector system on G_{15} yields a 15×30 sector–edge incidence matrix M with the expected row and column regularity.

7. The overlap matrix
$$Q := MM^{\mathsf{T}}$$
satisfies the distance-overlap formula
$$Q_{uv} = \begin{cases} 14, & d(u,v) = 0, \\ 9, & d(u,v) = 1, \\ 5, & d(u,v) = 2, \\ 4, & d(u,v) = 3, \end{cases}$$
equivalently
$$Q = 14D_0 + 9D_1 + 5D_2 + 4D_3.$$

8. The same matrix satisfies the polynomial incidence identity
$$Q = A^3 + 2A^2 + 2I.$$

9. After centering and normalization, the row vectors of M yield the stated three-angle spherical realization.

10. The lift
$$G_{30} \to G_{15}$$
defines a nontrivial $\mathbb{Z}_2$-valued cocycle.

A.2 Generation of the transport graph

The first computational step is the generation of the chamber transport graph from the local dodecahedral transport rules. Starting from the flag space of the combinatorial dodecahedron, the code applies the elementary transport moves and then passes to the opposite-edge quotient. The result is a graph G_{60} whose vertices are the 60 chamber states and whose edges record single-step admissible transport.

The following basic invariants are computed directly from the adjacency data:

- number of vertices;
- number of edges;
- degree sequence;
- connectedness;

- distance distribution from a root;
- graph diameter;
- triangle count;
- automorphism group order.

These checks confirm the invariant package recorded above.

A.3 External graph identification

To anchor the transport graph externally, the computed graph is compared against known graph encodings and census data. This includes canonical labeling and comparison with graph databases. The identification

$$G_{60} \cong \mathrm{AT4val}[60, 6]$$

is verified, with corresponding House of Graphs entry `Graph52002` [3, 2].

This step serves only to identify the transport graph externally. The main theorem of the paper does not depend on novelty of the graph itself, but on the transport-induced structure carried by its quotient core.

A.4 Verification of the quotient tower

The quotient tower

$$G_{60} \longrightarrow G_{30} \longrightarrow G_{15}$$

is verified by constructing the relevant equivalence relations and checking that the induced quotient graphs have the expected sizes and adjacency relations.

For each quotient step, the following are checked:

1. the equivalence classes have the expected cardinalities;
2. the quotient adjacency relation is well defined;
3. the quotient graph has the expected number of vertices and edges;
4. the quotient graph is independent of representative choice.

The deepest quotient is then compared against a canonical model of $L(\text{Petersen})$ using explicit graph-isomorphism testing.

A.5 Verification of the quotient core

The graph G_{15} is verified to satisfy

$$|V(G_{15})| = 15, \qquad |E(G_{15})| = 30, \qquad \deg(v) = 4,$$

and an explicit isomorphism

$$G_{15} \cong L(\text{Petersen})$$

is computed. Depending on the implementation, this is checked either by direct graph-isomorphism testing against a generated copy of $L(\text{Petersen})$ or by matching a canonical graph encoding.

This verification fixes the algebraic core used throughout the remainder of the paper.

A.6 Construction of the sector matrix

Once G_{15} is fixed, the transport-induced sector system is enumerated. For each vertex $v \in V(G_{15})$, the corresponding sector

$$S(v) \subseteq E(G_{15})$$

is listed explicitly. These sector supports are assembled into the sector–edge incidence matrix

$$M \in \{0,1\}^{15\times 30}.$$

The following are checked:

- each row of M agrees with the corresponding sector support;
- each row has weight 14;
- each column has weight 7;
- the matrix is compatible with the symmetries of the quotient core.

A.7 Verification of sector overlaps

From the incidence matrix M, the overlap matrix

$$Q := MM^{\mathsf{T}}$$

is computed directly. The entries are then compared against graph distance in G_{15}.

The verification checks that

$$Q_{uv} = |S(u) \cap S(v)|$$

takes the values

$$14, \quad 9, \quad 5, \quad 4$$

according as the distance $d(u,v)$ is

$$0, \quad 1, \quad 2, \quad 3.$$

This yields the distance expansion

$$Q = 14D_0 + 9D_1 + 5D_2 + 4D_3.$$

A.8 Verification of the polynomial identity

The adjacency matrix A of G_{15} is computed, and the polynomial matrix

$$A^3 + 2A^2 + 2I$$

is formed explicitly. This matrix is then compared entrywise with

$$Q = MM^{\mathsf{T}}.$$

The computational verification consists of the following steps:

1. compute the adjacency matrix A of G_{15};
2. compute $A^3 + 2A^2 + 2I$;
3. compute MM^{T};
4. check exact equality of the resulting 15×15 matrices.

This confirms the polynomial incidence identity.

A.9 Spectral consistency checks

As a consistency check, the spectrum of $G_{15} \cong L(\text{Petersen})$ is computed and verified to be

$$\{-2^{(5)},\, -1^{(4)},\, 2^{(5)},\, 4^{(1)}\}.$$

Applying the polynomial

$$p(x) = x^3 + 2x^2 + 2$$

to these eigenvalues yields

$$p(-2) = 2, \qquad p(-1) = 3, \qquad p(2) = 18, \qquad p(4) = 98.$$

The spectrum of $Q = MM^{\mathsf{T}}$ is then checked against the same multiset. This gives an independent spectral confirmation of the main identity.

A.10 Centering and spherical verification

The spherical realization is verified from the rows of M by computing

$$\mu = \frac{1}{15}\sum_{v} m_v, \qquad \widetilde{m}_v = m_v - \mu, \qquad x_v = \frac{\widetilde{m}_v}{\|\widetilde{m}_v\|}.$$

All pairwise inner products among the normalized vectors are then computed. The verification checks that for distinct vertices u, v,

$$\langle x_u, x_v \rangle = \begin{cases} \frac{37}{112}, & d(u,v) = 1, \\ -\frac{23}{112}, & d(u,v) = 2, \\ -\frac{38}{112}, & d(u,v) = 3. \end{cases}$$

Thus the induced Euclidean configuration is confirmed to be a three-angle spherical code controlled by graph distance in the core.

A.11 Signed lift and cocycle verification

The intermediate quotient

$$G_{30} \to G_{15}$$

is verified to be a signed 2-lift by examining how each edge of the base graph lifts to the double cover. Each edge is classified as either parallel or crossed, yielding the sign function

$$\sigma : E(G_{15}) \to \{\pm 1\}$$

and the associated cocycle

$$\varepsilon : E(G_{15}) \to \mathbb{Z}_2.$$

The following checks are performed:

- lift behavior is consistent across each base edge;
- cycle holonomy agrees with the parity sum of cocycle values;
- switching at vertices changes the cocycle by a coboundary;

- the switching class contains no trivial representative;
- a minimal-support representative is found in tree gauge.

These computations establish that the cocycle is nontrivial and independent of the quadratic overlap geometry.

A.12 Artifact organization

The finite data supporting the computations is organized through manuscript artifacts, including graph encodings, quotient maps, sector supports, matrix exports, spherical Gram data, signed-lift data, and cocycle verification files. The appendices that follow summarize these artifacts by role:

- Appendix B: local dodecahedral transport input;
- Appendix C: explicit sector and matrix data;
- Appendix D: signed-lift and cocycle data.

A.13 Summary

The computations recorded here verify every finite claim used in the paper: the construction of G_{60}, the quotient tower, the identification

$$G_{15} \cong L(\text{Petersen}),$$

the construction of M, the distance-overlap formula for

$$Q = MM^{\mathsf{T}},$$

the polynomial identity

$$Q = A^3 + 2A^2 + 2I,$$

the induced spherical realization, and the nontrivial signed-lift cocycle.

B Dodecahedral Labeling and Transport Rules

This appendix records the local combinatorial input of the transport construction. Its purpose is to make the finite data underlying the chamber graph G_{60} explicit and reproducible.

B.1 The combinatorial dodecahedron

Let $\mathcal{D}$ denote the abstract combinatorial dodecahedron. Its incidence structure consists of

$$|V(\mathcal{D})| = 20, \qquad |E(\mathcal{D})| = 30, \qquad |F(\mathcal{D})| = 12.$$

A *flag* of $\mathcal{D}$ is a mutually incident triple

$$(v, e, f),$$

where v lies on e and e lies on f. The total number of flags is

$$120.$$

The transport construction begins on this flag space. The 60-vertex chamber graph is obtained only after quotienting by the opposite-edge identification described below.

B.2 Labeling conventions

For explicit computation, we fix label sets

$$V(\mathcal{D}) = \{v_1, \dots, v_{20}\}, \qquad E(\mathcal{D}) = \{e_1, \dots, e_{30}\}, \qquad F(\mathcal{D}) = \{f_1, \dots, f_{12}\}.$$

The implementation uses concrete incidence tables with these label classes. The essential requirement is that the same labeling be used consistently throughout the transport tables, quotient maps, graph encodings, and derived sector data.

B.3 Local incidence data

For each edge $e \in E(\mathcal{D})$, let

$$\partial_V(e) = \{v, v'\}$$

be its incident vertex pair, and let

$$\partial_F(e) = \{f, f'\}$$

be its incident face pair.

For each face $f \in F(\mathcal{D})$, let

$$\partial_E(f) = \{e_{f,1}, e_{f,2}, e_{f,3}, e_{f,4}, e_{f,5}\}$$

be its cyclically ordered boundary edges.

For each vertex $v \in V(\mathcal{D})$, let

$$\partial_E(v) = \{e_{v,1}, e_{v,2}, e_{v,3}\}$$

be the three incident edges, listed in the cyclic order induced by the local combinatorics of the dodecahedron.

These cyclic orders determine the transport rules.

B.4 Indexed flag set

For computational purposes, the 120 flags are indexed by a fixed ordering

$$\mathcal{C} = \{c_1, \dots, c_{120}\},$$

where each c_i is a flag (v, e, f). This indexed flag list is the level on which the transport rules are implemented.

A representative example is:

$$c_i = (v_a, e_b, f_c),$$

with

$$v_a \in \partial_V(e_b), \qquad e_b \in \partial_E(f_c).$$

The full indexed list is stored in the artifact set associated with the manuscript.

B.5 Face transport

The first transport type moves within a common face while changing the incident edge and vertex according to the chosen cyclic rule.

Definition B.1 (Face transport)**.** *Let* (v, e, f) *be a flag. The* face transport *move sends*

$$(v, e, f) \mapsto (v'_f, e'_f, f),$$

where e'_f *is the next admissible edge of the face* f *under the fixed cyclic order, and* v'_f *is the endpoint of* e'_f *selected by the transport orientation.*

At the conceptual level, face transport keeps the face fixed and advances the chamber along the incidence structure internal to that face.

A representative example has the form

$$(v_a, e_b, f_c) \mapsto (v_{a'}, e_{b'}, f_c),$$

where $e_{b'}$ is the next face-admissible edge in the chosen cyclic order on $\partial_E(f_c)$.

B.6 Vertex transport

The second transport type moves within a common vertex while changing the incident edge and face according to the chosen cyclic rule.

Definition B.2 (Vertex transport)**.** *Let* (v, e, f) *be a flag. The* vertex transport *move sends*

$$(v, e, f) \mapsto (v, e'_v, f'_v),$$

where e'_v *is the next admissible edge at the vertex* v *under the fixed cyclic order, and* f'_v *is the face incident with* e'_v *determined by the transport orientation.*

At the conceptual level, vertex transport keeps the vertex fixed and advances the chamber through the local incidence structure around that vertex.

A representative example has the form

$$(v_a, e_b, f_c) \mapsto (v_a, e_{b''}, f_{c'}),$$

where $e_{b''}$ is the next edge in the cyclic ordering of $\partial_E(v_a)$.

B.7 Opposite-edge identification

The raw 120-flag system contains a built-in twofold redundancy. Opposite local configurations across an edge determine equivalent chamber states for the purposes of the transport construction.

Definition B.3 (Opposite-edge equivalence)**.** *Two flags are* opposite-edge equivalent *if they represent opposite local states across the same dodecahedral edge under the transport identification used in the construction.*

This equivalence relation partitions the flag set into 60 classes.

Definition B.4 (Chamber states)**.** *The vertices of the transport graph* G_{60} *are the* 60 *equivalence classes of flags under opposite-edge identification.*

B.8 Construction of the chamber graph

Once opposite-edge classes are fixed, the transport graph is obtained by declaring two chamber states adjacent whenever one is obtained from the other by a single admissible transport move at the flag level.

Equivalently, if $[c]$ denotes the equivalence class of a flag c, then

$$[c] \sim [c'] \quad \Longleftrightarrow \quad \text{there exists an admissible transport move } c \mapsto c'.$$

This produces the 60-vertex chamber graph

$$G_{60}.$$

B.9 Stored construction artifacts

The explicit finite realization of the construction is recorded in the artifact set through data of the following kinds:

- vertex–edge incidence tables for $\mathcal{D}$;
- face boundary tables for $\mathcal{D}$;
- the indexed flag list $\{c_1, \ldots, c_{120}\}$;
- face-transport lookup tables on flags;
- vertex-transport lookup tables on flags;
- the opposite-edge equivalence-class table producing the 60 chamber states.

These files provide the exact finite input from which the chamber graph and the subsequent quotient structures are computed.

B.10 Summary

This appendix records the local combinatorial data behind the transport construction: the labeled dodecahedron, its indexed flag set, the face and vertex transport rules, and the opposite-edge quotient producing the chamber graph G_{60}. The main body of the paper uses these ingredients abstractly; the present appendix records them in the concrete form needed for reproducibility.

C Explicit Sector Data

This appendix records the finite sector data underlying the matrix

$$M \in \{0,1\}^{15\times 30}$$

and the Thalean quadratic

$$Q = MM^{\mathsf{T}}.$$

Its purpose is to expose the raw combinatorial information behind the incidence and spherical claims in the main body of the paper.

C.1 The core graph and its indexing

Throughout this appendix, let

$$G_{15} \cong L(\text{Petersen})$$

denote the core graph. Fix an ordering of its vertex set

$$V(G_{15}) = \{u_1, \ldots, u_{15}\}$$

and an ordering of its edge set

$$E(G_{15}) = \{a_1, \ldots, a_{30}\}.$$

This indexing is the one used for the sector lists, the incidence matrix M, the overlap matrix Q, and the centered and normalized vector data. The same ordering is also used in Appendix D for the signed-lift and cocycle layer.

C.2 Sector lists

For each core vertex u_i, the Thalean construction determines a sector

$$S(u_i) \subseteq E(G_{15}).$$

These sectors may be recorded explicitly as edge subsets:

$$S(u_i) = \{a_{i_1}, a_{i_2}, \ldots, a_{i_k}\}.$$

The full data is most naturally stored as a machine-readable list of the fifteen sector supports

$$S(u_1), \ldots, S(u_{15}),$$

with each support written in terms of the edge indexing $\{a_1, \ldots, a_{30}\}$. In the typeset presentation, these supports may be displayed either one per line or in a compact table.

A template form is

$$\begin{aligned} S(u_1) &= \{\cdots\}, \\ S(u_2) &= \{\cdots\}, \\ &\vdots \\ S(u_{15}) &= \{\cdots\}. \end{aligned}$$

These supports are the defining data from which the matrix M is constructed.

C.3 The sector–edge incidence matrix

With the row and column orderings fixed, the sector–edge incidence matrix is

$$M = (M_{ij}) \in \{0,1\}^{15\times 30},$$

where

$$M_{ij} = \begin{cases} 1, & a_j \in S(u_i), \\ 0, & a_j \notin S(u_i). \end{cases}$$

A convenient way to record M is by row support:

$$\operatorname{supp}(m_i) = \{j : M_{ij} = 1\},$$

where m_i is the i-th row of M. This representation is equivalent to the full binary matrix and is usually easier to inspect by eye.

The matrix M may therefore be presented in one of three equivalent forms:

- as a displayed 15×30 binary matrix;
- as a row-by-row support list;
- as a machine-readable artifact, with the support list summarized in the PDF.

C.4 Row and column regularity

The matrix M satisfies strong regularity conditions. Each row has the same weight, since each sector has the same cardinality, and each column has the same weight, since each core edge lies in the same number of sectors. In the present construction,

$$|S(u_i)| = 14 \quad \text{for all } i,$$

and

$$|\{i : a_j \in S(u_i)\}| = 7 \quad \text{for all } j.$$

A compact summary is therefore

quantity	value
number of sectors	15
number of core edges	30
row weight	14
column weight	7

This uniformity is one of the main reasons the overlap matrix admits such a rigid polynomial description.

C.5 The overlap matrix

From the sector matrix we form

$$Q := MM^{\mathsf{T}}.$$

The entries of Q are

$$Q_{ij} = |S(u_i) \cap S(u_j)|.$$

The main text proves that the overlap matrix depends only on graph distance in G_{15}, with distance-class expansion

$$Q = 14D_0 + 9D_1 + 5D_2 + 4D_3.$$

Equivalently,

$d(u_i, u_j)$	Q_{ij}
0	14
1	9
2	5
3	4

Thus the full 15×15 matrix Q can be reconstructed from the distance matrix of the core graph.

C.6 Distance matrix of the core

For completeness, one may also record the distance matrix

$$D = (d(u_i, u_j))_{1 \le i,j \le 15}$$

of the core graph. Since the overlap matrix Q depends only on distance, the pair (D, Q) provides a transparent finite summary of the entire quadratic sector geometry.

A compact presentation of the core data may therefore include:

1. the adjacency matrix A of G_{15};
2. the distance matrix D;
3. the overlap matrix Q.

C.7 Centered sector vectors

Let $m_1, \ldots, m_{15} \in \mathbb{R}^{30}$ denote the rows of M. Define the mean sector vector

$$\mu = \frac{1}{15} \sum_{i=1}^{15} m_i$$

and the centered vectors

$$\widetilde{m}_i = m_i - \mu.$$

The centered configuration is the Euclidean form of the nontrivial part of the sector geometry. In practice, the centered data may be recorded either as the vectors $\widetilde{m}_i$ themselves or, more economically, through the centered Gram matrix

$$\widetilde{Q}_{ij} = \langle \widetilde{m}_i, \widetilde{m}_j \rangle.$$

Since the main text uses these vectors only through their inner products, the centered Gram matrix is usually the most compact artifact to preserve.

C.8 Normalized spherical data

To obtain the spherical realization, each centered vector is normalized:

$$x_i = \frac{\widetilde{m}_i}{\|\widetilde{m}_i\|}.$$

The normalized inner products satisfy the three-angle relation proved in the main text. For $i \neq j$,

$$\langle x_i, x_j \rangle \in \left\{ \frac{37}{112}, -\frac{23}{112}, -\frac{38}{112} \right\},$$

with the value determined by graph distance in G_{15}. A compact verification table is therefore

$d(u_i, u_j)$	$\langle x_i, x_j \rangle$
1	$\frac{37}{112}$
2	$-\frac{23}{112}$
3	$-\frac{38}{112}$

This table is the normalized spherical analogue of the overlap summary for Q.

C.9 Machine-readable artifacts

The full sector data is naturally organized through machine-readable files. A typical artifact set includes:

- `sector_sets.json`,
- `sector_incidence_matrix.csv`,
- `overlap_matrix.csv`,
- `centered_gram_matrix.csv`,
- `normalized_gram_matrix.csv`.

Together, these files provide the explicit finite realization of the sector system, the Thalean quadratic, and the induced spherical configuration.

C.10 Summary

This appendix contains the raw sector data behind the main incidence and spherical results: the sector sets, the matrix M, the overlap matrix

$$Q = MM^{\mathsf{T}},$$

and the centered and normalized geometric data derived from M. The purpose of these tables and artifacts is not to advance the conceptual story, but to make the paper's finite claims directly checkable from explicit data.

D Signed-Lift and Cocycle Data

This appendix records the finite data underlying the signed lift

$$G_{30} \to G_{15}$$

and the associated $\mathbb{Z}_2$-valued cocycle

$$\varepsilon : E(G_{15}) \to \mathbb{Z}_2.$$

Its purpose is to make the cocycle layer of the Thalean construction directly inspectable. The main text gives the conceptual and cohomological interpretation; the present appendix collects the concrete edge-level data used to verify it.

D.1 Base graph indexing

Let

$$G_{15} \cong L(\text{Petersen})$$

be the core graph, with vertex set

$$V(G_{15}) = \{u_1, \dots, u_{15}\}$$

and edge set

$$E(G_{15}) = \{a_1, \dots, a_{30}\}.$$

This indexing is the one used throughout the finite data of the paper. The same ordering of edges is used for the adjacency matrix of G_{15}, the sector–edge incidence matrix M, the cocycle ε, and the switching-class representatives and holonomy tables recorded below.

D.2 The signed lift

The intermediate quotient

$$G_{30} \to G_{15}$$

is a 2-sheeted lift. Each vertex $u_i \in V(G_{15})$ has a fiber of size 2, which we denote

$$u_i^+, \quad u_i^-.$$

Each edge $a_j = \{u_r, u_s\} \in E(G_{15})$ lifts in one of two ways:

- *parallel*:
$$u_r^+ \sim u_s^+, \qquad u_r^- \sim u_s^-;$$
- *crossed*:
$$u_r^+ \sim u_s^-, \qquad u_r^- \sim u_s^+.$$

The lift type of each base edge is recorded explicitly in the machine-readable artifact set for the manuscript. In tabular form, the lift data is naturally organized as

base edge	endpoints in G_{15}	lift type
a_1	$\{u_{r_1}, u_{s_1}\}$	parallel/crossed
a_2	$\{u_{r_2}, u_{s_2}\}$	parallel/crossed
$\vdots$	$\vdots$	$\vdots$
a_{30}	$\{u_{r_{30}}, u_{s_{30}}\}$	parallel/crossed

D.3 Sign function and cocycle

The signed lift is equivalently encoded by a sign function

$$\sigma : E(G_{15}) \to \{\pm 1\},$$

where

$$\sigma(a_j) = \begin{cases} +1, & \text{if } a_j \text{ lifts in parallel,} \\ -1, & \text{if } a_j \text{ lifts crossed.} \end{cases}$$

Passing to $\mathbb{Z}_2$, define

$$\varepsilon(a_j) = \begin{cases} 0, & \sigma(a_j) = +1, \\ 1, & \sigma(a_j) = -1. \end{cases}$$

Thus the cocycle is recorded as a vector

$$(\varepsilon(a_1), \ldots, \varepsilon(a_{30})) \in \mathbb{Z}_2^{30}.$$

This vector is the basic finite encoding of the transport memory carried by the signed lift. In the typeset presentation it may be given as a compact table, while the complete edge-by-edge data is preserved in the corresponding artifact files.

D.4 Switching equivalence

The cocycle is defined intrinsically only up to switching at vertices. A switching function

$$\phi : V(G_{15}) \to \mathbb{Z}_2$$

acts on edge values by

$$\varepsilon'(uv) = \varepsilon(uv) + \phi(u) + \phi(v) \pmod 2.$$

Thus two cocycle representatives are considered equivalent if they differ by such a vertex switch. The switching class of the cocycle is the invariant object attached to the lift.

The artifact set records:

- the raw cocycle representative extracted from the lift;
- the switched representatives used in computation;
- the preferred representative used in the manuscript.

D.5 Tree gauge

A convenient normal form is obtained by fixing a spanning tree

$$T \subseteq G_{15}$$

and switching so that every tree edge has cocycle value 0. This is the tree gauge used in the cocycle analysis.

In tree gauge, one has

$$\varepsilon(e) = 0 \qquad \text{for all } e \in E(T),$$

and the support of the cocycle is concentrated on the non-tree edges. A compact description is therefore

$$\mathrm{supp}(\varepsilon_T) = \{a_{j_1}, a_{j_2}, \ldots, a_{j_m}\},$$

where ε_T denotes the tree-gauge representative.

This gauge makes the nontriviality of the cocycle directly visible in terms of fundamental cycles.

D.6 Fundamental cycles

Once a spanning tree T is fixed, each non-tree edge $a_j \notin E(T)$ determines a unique fundamental cycle

$$C_j \subseteq G_{15}.$$

In tree gauge, the holonomy parity of C_j is exactly

$$\omega(C_j) = \varepsilon_T(a_j).$$

Thus the support of the cocycle is directly visible in the fundamental-cycle basis. The holonomy data is most naturally summarized in a table of the form

non-tree edge	fundamental cycle	$\omega(C_j)$
a_{j_1}	C_{j_1}	0/1
a_{j_2}	C_{j_2}	0/1
$\vdots$	$\vdots$	$\vdots$

D.7 Minimal support representative

Up to switching equivalence, the Thalean cocycle admits a representative of minimal support size 6. In other words, there is a representative $\varepsilon_{\min}$ in the switching class of ε such that

$$|\mathrm{supp}(\varepsilon_{\min})| = 6.$$

The support set of this representative is recorded explicitly in the machine-readable artifact set. This shows that the cocycle is sparse in a suitable gauge, but not eliminable.

D.8 Holonomy table

To make the cocycle directly testable, it is useful to record sample cycles and their holonomy values. The natural format is

cycle C	$\omega(C) = \sum_{e \in C} \varepsilon(e) \pmod 2$
C_1	0/1
C_2	0/1
$\vdots$	$\vdots$

The most informative examples are:

- the fundamental cycles associated with the chosen tree gauge;
- short cycles in G_{15};
- any distinguished cycles used to witness nontrivial holonomy.

Together, these computations provide direct evidence that the cocycle is not a coboundary.

D.9 Cohomological verification

The nontriviality claim

$$[\varepsilon] \neq 0 \in H^1(G_{15}, \mathbb{Z}_2)$$

is verified in finite form in several equivalent ways:

- by exhibiting a cycle with odd holonomy;
- by recording a tree-gauge representative with nonempty support;
- by checking that ε does not lie in the coboundary subspace.

A linear-algebra formulation is particularly convenient. Let B be the vertex-edge incidence matrix of G_{15} over $\mathbb{Z}_2$. Then ε is a coboundary if and only if it lies in the column space of B^{T}. The computational verification shows that this is not the case.

D.10 Failure of local determination

The body of the paper proves that the cocycle cannot be recovered from any finite-radius automorphism-invariant local rule on edges of G_{15}. The finite data supporting this claim is simple:

- G_{15} is edge-transitive;
- rooted local edge neighborhoods of fixed radius are isomorphic;
- the cocycle nevertheless takes both values 0 and 1.

In particular, there exist edges $a_i, a_j \in E(G_{15})$ such that

$$\varepsilon(a_i) \neq \varepsilon(a_j),$$

even though the corresponding rooted local edge neighborhoods are isomorphic. This confirms that the cocycle is not determined by local edge data on the core graph.

D.11 Machine-readable artifacts

The signed-lift and cocycle data is organized through discrete artifact files, including:

- `signed_lift_edges.json`,
- `cocycle_vector.csv`,
- `tree_gauge_representative.csv`,
- `fundamental_cycle_holonomy.csv`,
- `cocycle_linear_check.txt`.

Together, these files provide the full explicit finite realization of the signed-lift layer used in the paper.

D.12 Summary

This appendix records the explicit data behind the cocycle layer of the Thalean construction: the signed 2-lift

$$G_{30} \to G_{15},$$

the edge signing σ, the $\mathbb{Z}_2$-valued cocycle ε, switching-class representatives, support sets, and holonomy computations. Its purpose is to make the paper's topological claims as directly checkable as the incidence and spherical claims recorded in the previous appendix.

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